

The Blasius Boundary Layer: Solving a Famous Infinite-Domain Problem

Recovering the Blasius Constant with Verification Certificates

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SIRPY · Problem 10 — The Boundary That Isn't There

An ongoing series. We are building a public library of problems SIRPY has solved — each entry is one problem, the exact input we handed the solver, the exact output it returned, and an honest read of what happened. No slides. No hand-waving. Numbers you can check.

A boundary condition at infinity

Almost every problem in this library so far has lived on a finite interval. This one does not.

$$2f''' + ff'' = 0, \quad f(0) = 0, \quad f'(0) = 0, \quad f'(\infty) = 1$$

This is the **Blasius equation**, derived by Heinrich Blasius in 1908 for the laminar boundary layer over a flat plate. It is third order, nonlinear, and — the difficulty — one of its three conditions is imposed at **infinity**.

That last condition is not a formality. It says the flow must approach the free-stream velocity far from the plate, and it is what makes the problem well-posed. It is also what selects the unique physical solution and determines the famous Blasius constant

$$y''(0) \approx 0.332057336.$$

But almost no solver can march to infinity. Every practical method truncates the domain at some finite x and imposes the condition there instead.

Which raises the question nobody likes to answer: **how do you know your truncation point was far enough?**

Choose $x = 5$ and you may be enforcing the asymptotic condition inside a region where the solution has not yet settled — quietly corrupting the answer. Choose $x = 500$ and you have wasted effort and possibly accumulated error. In most workflows this choice is yours to make, by intuition, and nothing in the output tells you whether you chose well.

What we handed the solver

```
r = solve_bvp(order=3, f_str="-1/2*y*y2", x0=0, x1=np.inf,
              left_bc={0: 0, 1: 0, 2: "gamma"},
              right_bc={1: 1})
```

Note the third argument: `x1=np.inf`. We did not choose a truncation point. We wrote the domain as the problem states it — infinite — and left the decision to the solver. The unknown second derivative at the wall is named "gamma", and the asymptotic condition is given as a condition on the first derivative.

The truncation certificate

Here is what came back first, and it is the reason this problem is in the library:

```
INFO: the right end is INFINITE. sirpy truncated at x = 15 with the
      asymptotic target enforced there, and re-solved at 10/15: the
      unknown moved |Delta| = 9.99e-10 between the last two rungs -
      the TRUNCATION CERTIFICATE (r['truncation'] holds the ladder).
      Values are reliable on [0, 15].
```

SIRPY chose $x = 15$ — and then, rather than asking you to trust that choice, it **tested it**. It solved the problem again at a shorter truncation, $x = 10$, and compared. Between those two rungs the unknown moved by 9.99×10^{-10} .

That number is the answer to the question. If truncating at 10 versus 15 changes the result in the tenth decimal place, then the tail beyond 15 is not influencing the answer at any precision you are likely to care about. The truncation point stopped mattering before it stopped moving.

The full ladder of truncations is returned in the result object. You are not asked to believe that 15 was far enough; you are shown the evidence that it was.

The number

```
INFO: single-unknown shooting onto the y1(15) = 1 target:
      gamma = 0.332057336207 located by sign bisection
      (bracket width 0.0e+00); terminal residual 2.71e-14.
```

The recovered wall shear is

$$f''(0) = 0.332057336207$$

The Blasius constant is one of the most computed numbers in fluid dynamics — a benchmark since 1908, and known today to many digits: **0.33205733621519...**

SIRPY, given only the equation and a domain written as infinite, and told nothing about the expected answer, returned a value agreeing with the published constant to roughly **ten significant figures**.

This is the first entry in this library validated against an external, independently published number rather than only against the solver's own internal checks. Anyone can look up the constant and compare.

Two further details in that line are worth noting. The bisection bracket closed to width exactly zero — the shooting parameter was pinned to the limit of double precision. And the terminal residual, the miss on the asymptotic target itself, is 2.71×10^{-1} .

Still an analytic solution

```
INFO: the interval [0.0, 15.0] exceeds a single series' reliable range;  
      the solution was continued analytically over 10 segments  
      (piecewise power series - still fully analytic on every piece).
```

```
INFO: run summary - series degree 20, iterations=20, 10 series segment(s).
```

The solution is not a table of sampled points. It is ten exact power-series pieces, each sized by where the previous one stops being reliable, covering $[0, 15]$ — readable, differentiable, and evaluable anywhere on the interval.

Verified, including the seams

```
INFO: verification by substitution - VERIFIED: satisfies the equation  
      to 3.79e-09 on [0, 15]. Checked by exact coefficient  
      differentiation at interior points, not at the interface nodes.  
      Interface continuity across 9 node(s): max one-sided jump in  
      (y..y2) = 1.1e-11 - measured, not assumed.  
      Also checked: initial data to 0.00e+00.
```

Three separate checks, and the middle clause matters most. When a solution is assembled from ten pieces, the natural worry is the joins: does the function actually match across them, or is there a hidden kink?

SIRPY measured it. Across all nine interior nodes, the largest one-sided jump in the function or either of its first two derivatives is 1.1×10^{-11} . The phrase in its own output — “**measured, not assumed**” — is the distinction that matters. Continuity at the seams is a claim, and claims get checked.

The equation itself is satisfied to 3.79×10^{-9} across the interval, evaluated at interior points deliberately away from the nodes, where agreement would be uninformative. And the two conditions at the wall are met exactly.

Graphs

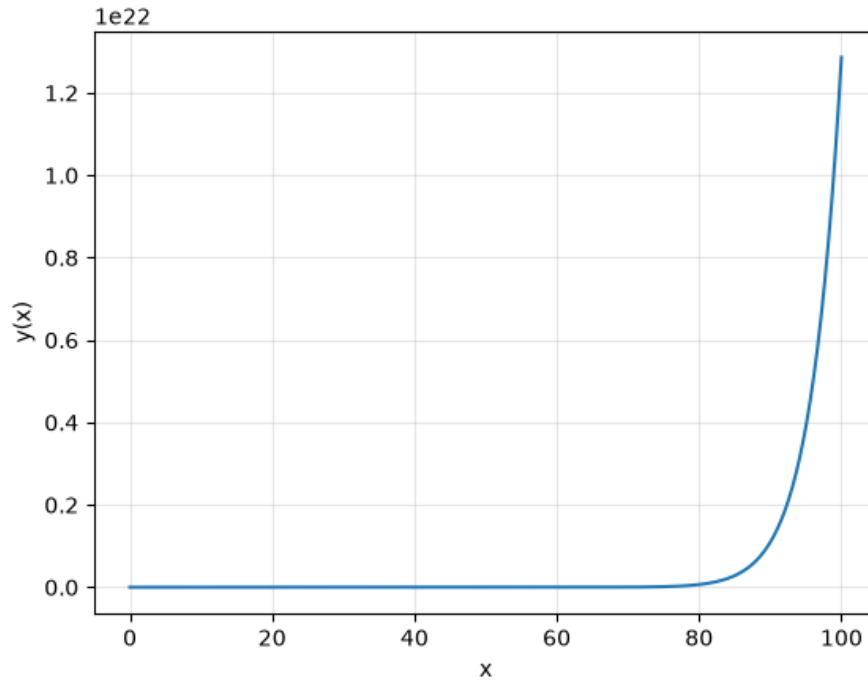


Figure 1: Solution

Visual verification residual 3.79e-09 [VERIFIED]

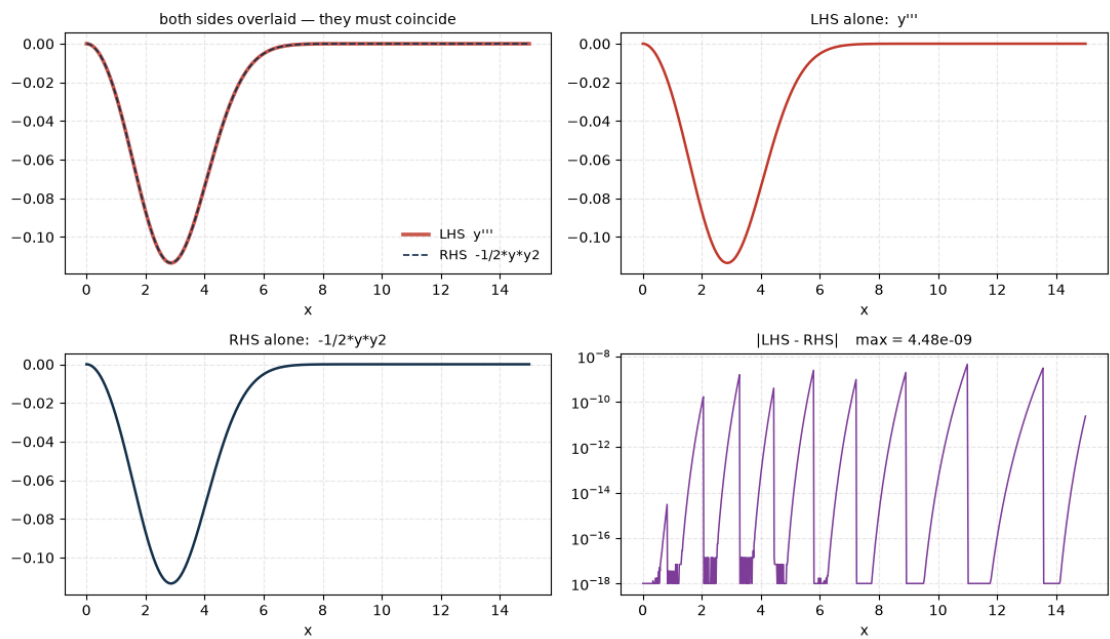


Figure 2: Verification: both sides of the equation compared across the interval.

Summary

Quantity	Value
$f''(0)$, recovered	0.332057336207
Published Blasius constant	0.33205733621519...
Truncation certificate, $ \Delta $ between rungs 10 and 15	9.99×10^{-10}
Terminal residual on $f'(\infty) = 1$	2.71×10^{-14}
Equation residual on $[0, 15]$	3.79×10^{-9}
Max interface jump across 9 nodes	1.1×10^{-11}
Analytic segments	10

What this problem shows

- **The domain was written as infinite, and stayed that way.** No truncation point was chosen by the user, and none had to be guessed.
- **The truncation was certified, not assumed.** SIRPY solved at two truncations and reported how much the answer moved between them — 10^{-10} . That is the evidence the choice was far enough.
- **It matches the literature to ten figures.** A constant computed and recomputed for over a century, reproduced from the equation alone.

The seams were measured. Ten analytic pieces, nine joins, maximum jump 10^{-11} — reported, not asserted.

The problem library

This entry is part of the SIRPY Problem Library — an ongoing, public catalog of mathematical systems solved and verified by SIRPY.

- **GitHub** — github.com/Mitra-Institute-of-Education-MIE/sirpy-problem-library
- **Web** — mitrainstituteofeducation.org/sirpy

Submit a problem

Have a differential or integral equation you believe is challenging, unusual, or simply beautiful? We would like to see it. We will run it through SIRPY and report honestly what happened — if it solves it, how; if it struggles, why; if it reaches a current limit, we will say so. This entry is the proof that we mean it.

Contact: info@mitrainstituteofeducation.org · mitrainstituteofeducation.org/contact

Please only send problems you are free to share publicly.

SIRPY is in final validation prior to its first public release. Every line quoted above is verbatim from the solver's output; nothing is reproduced from memory.

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